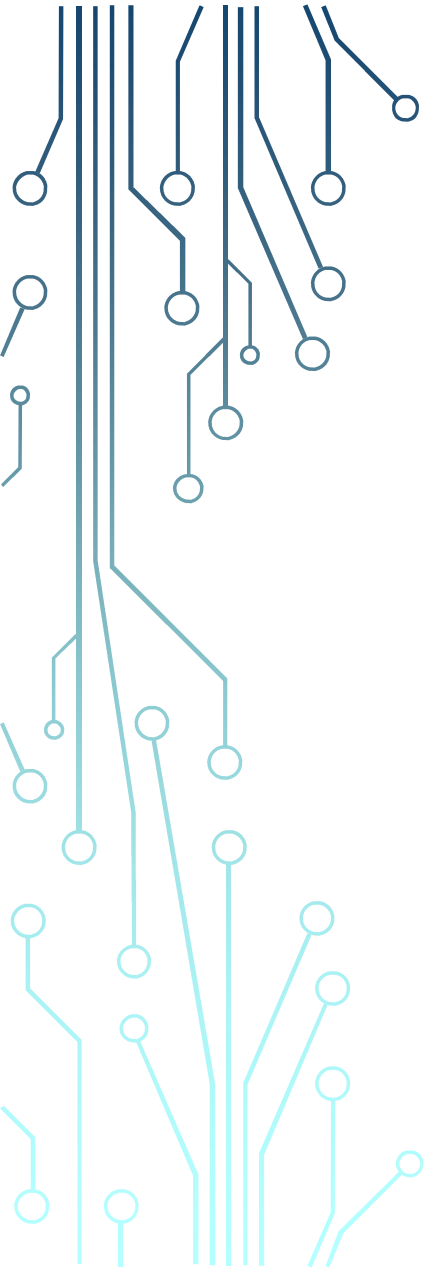
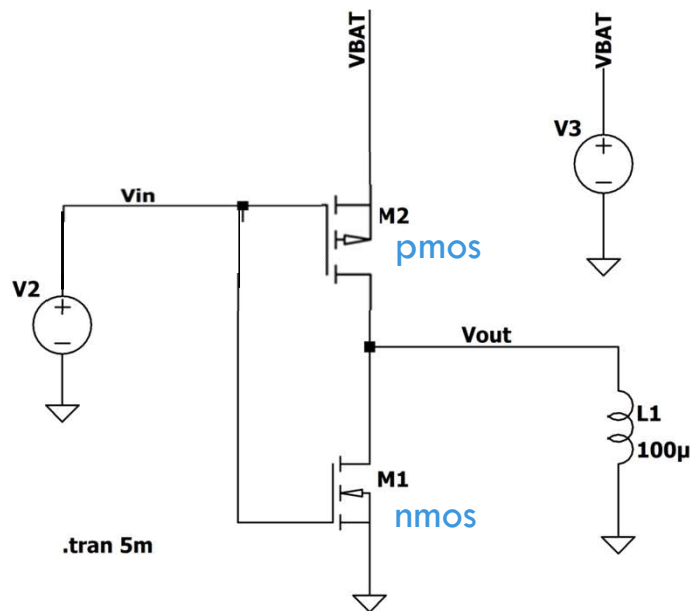




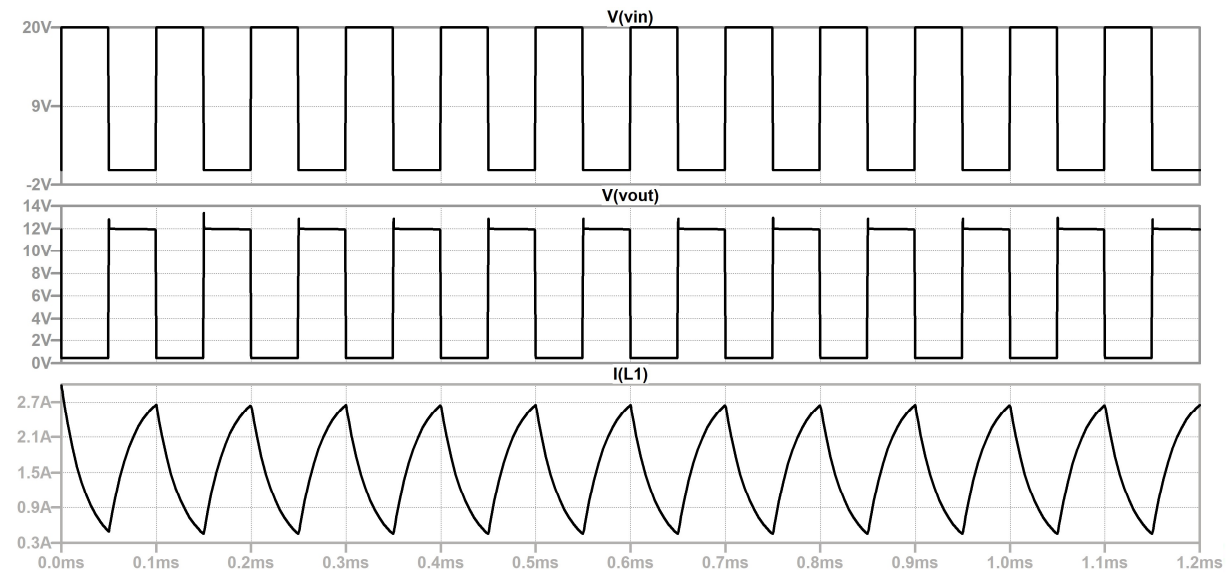
Bootstrap, Charge-Pump Circuits and Isolated DCDC Converters for Gate Drivers

Bootstrap Circuits

- Half-bridge circuits are widely used in power electronics: DCDC converters, motor drivers, rectifiers...
- It is important to have the right correct drain-source and gate-source voltages, otherwise, the half-bridge circuit will not work properly.
- Why don't we use this nmos/pmos combination (inverter)? Compared with nmos, pmos transistors have a higher $R_{DS,on}$, are slower and therefore have a higher conduction and switching loss.



works without
any problems!



$$V_L = L \frac{di_L(t)}{dt} \Rightarrow i_L = \frac{1}{L} \int V_L(t) \cdot dt$$



Bootstrap Circuits

- If losses and speed (high current and high switching frequency) are important, pmos/nmos complementary circuit is not ideal due to higher losses.
 - pmos/nmos circuit is not symmetrical leading to a distortion at the output signal.
 - nmos / nmos would be the first choice for high current and high-speed applications: motor drives, DCDC converters, rectifiers etc.
- 
- 
- 

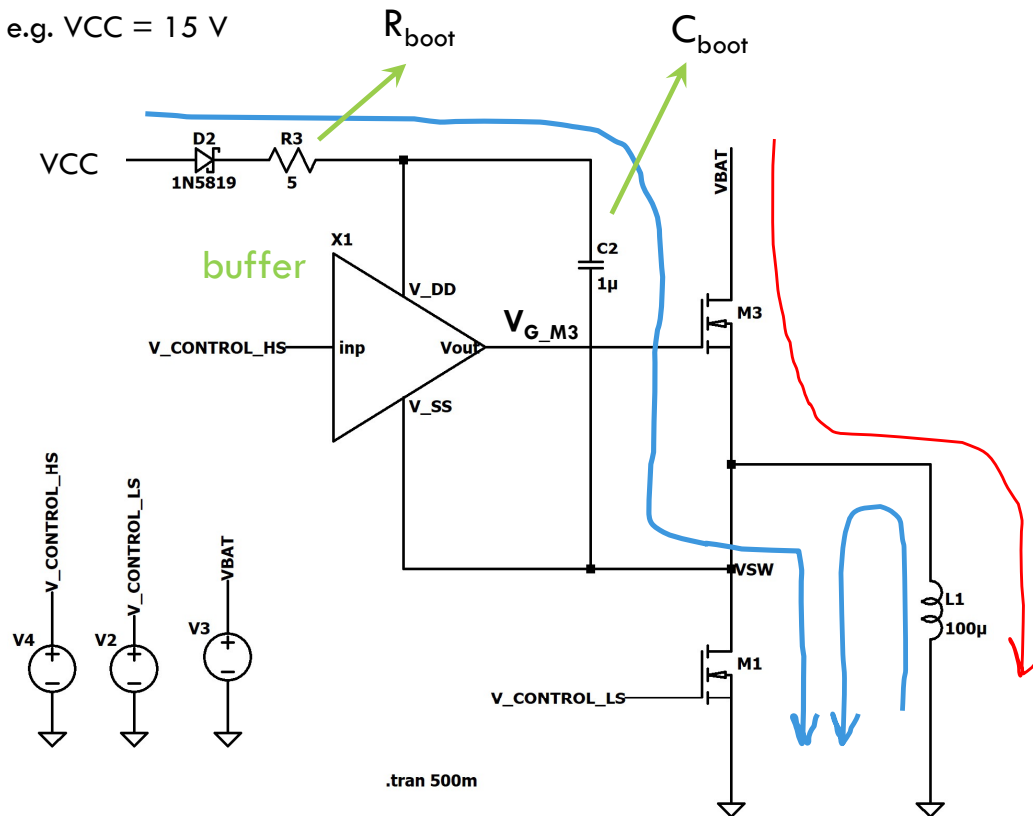
How to generate 22 V, if the maximum voltage in the system is 12 V (V_{BAT})?

Connect a voltage source between gate and source of M3, which generates a PWM wave.

- How can such a voltage source be implemented in a real-world application? → the ground reference of PWM voltage source would have to be $V_{SW}!!!$

Bootstrap Circuits

e.g. $V_{CC} = 15\text{ V}$



VSW becomes the ground of the **buffer**!

— : Low-Side Switch ON

— : High-Side Switch ON

1) When the low-side switch M1 is on, M1 drains current from the inductive load L and a current flows from VBAT through D2, R3, C2 and M1. R3 is a current limiting resistor.

2) V_CONTROL_HS and V_CONTROL_LS are generated by the microcontroller and complementary signals.

3) When the high-side switch M3 is on, M3 injects current into the inductive load L. M3 gate signal V_CONTROL_HS is buffered by an inverter. When this buffer drives M3, it discharges the capacitor C2. In other words, the buffer is supplied by C2.

Bootstrap Circuits

Total charge drawn from the bootstrap capacitor

$$Q_{total} = Q_g + Q_{driver}$$

Q_g :

Gate charge of the high-side MOSFET. This is the charge required to charge the MOSFET gate capacitances (C_{gs} and C_{gd}) and raise the gate to the required gate-source voltage.

Q_{driver} :

Charge consumed by the internal circuitry of the high-side gate driver while the high-side MOSFET remains ON.

Driver charge:

$$Q_{driver} = I_{driver} \cdot t_{ON}$$

Therefore, the total charge becomes

$$Q_{total} = Q_g + I_{driver} \cdot t_{ON}$$

Capacitor charge relation

$$Q = C \cdot \Delta V$$

Bootstrap Circuits

Voltage drop across the bootstrap capacitor

$$\Delta V_{BS} = \frac{Q_{total}}{C_{boot}}$$

Bootstrap capacitor sizing condition

$$C_{boot} \geq \frac{Q_{total}}{\Delta V_{BS}}$$

Maximum high-side ON time

$$C_{boot} \cdot \Delta V_{BS} = Q_g + I_{driver} \cdot t_{ON,max}$$

Solving for the maximum ON time

$$t_{ON,max} = \frac{C_{boot} \cdot \Delta V_{BS} - Q_g}{I_{driver}}$$

Capacitor Charging Phase

Minimum OFF time required to recharge the bootstrap capacitor

95%

99%

$$t_{OFF,min} \approx 3(R_{boot}C_{boot}) \quad \text{to} \quad 5(R_{boot}C_{boot})$$

Switching period

$$T = \frac{1}{f_{sw}}$$

Duty cycle definition

$$D = \frac{t_{ON}}{T}$$

$$D_{max} = \frac{t_{ON,max}}{T} \quad \text{and} \quad D_{max} = 1 - \frac{t_{OFF,min}}{T}$$

This condition ensures that sufficient OFF time exists for the bootstrap capacitor to recharge through the bootstrap diode.

Capacitor Discharge Phase

Bootstrap Circuits - Example

MOSFET total gate charge: $Q_g = 35 \text{ nC}$

High-side driver quiescent current: $I_{\text{driver}} = 0.5 \text{ mA}$

Bootstrap capacitor & resistor: $C_{\text{boot}} = 10 \text{ nF}$, $R_{\text{boot}} = 5 \Omega$

Allowable bootstrap voltage **drop**: $\Delta V_{\text{BS}} = 10 \text{ V}$

Switching frequency: $f_{\text{sw}} = 7.5 \text{ kHz}$

Maximum high-side ON time:

$$t_{\text{ON,max}} = (C_{\text{boot}} \cdot \Delta V_{\text{BS}} - Q_g) / I_{\text{driver}}$$

First compute the charge available from the capacitor

$$= 10 \text{ nF} \times 10 \text{ V}$$

$$= 100 \text{ nC}$$

Now compute the maximum ON time

$$t_{\text{ON,max}} = (100 \text{ nC} - 35 \text{ nC}) / 0.5 \text{ mA}$$

$$t_{\text{ON,max}} = 65 \text{ nC} / 0.5 \text{ mA} = 130 \mu\text{s}$$

$$T = 1 / 7.5 \text{ kHz} = 133.3 \mu\text{s}$$

$$D_{\text{max}} = 130 \mu\text{s} / 133.33 \mu\text{s} = 97.5\%$$

So if the switching frequency is 7.5 kHz, the bootstrap charge limitation allows the MOSFET to stay ON for 130 μs out of a 133.33 μs period, corresponding to a maximum duty cycle of about 97.5%.

However, we must still check the bootstrap recharge condition:

$$t_{\text{OFF}} = 133.3 \mu\text{s} - 130 \mu\text{s} = 3.33 \mu\text{s}$$

If the required minimum recharge time was

$$t_{\text{OFF,min}} \approx 5 \times (10 \text{ nF} \times 5) = 0.25 \mu\text{s}$$

Bootstrap Circuits - Example

$$0.25 \mu\text{s} < 3.33 \mu\text{s}$$

which means the bootstrap capacitor can recharge properly.

From the recharge constraint:

$$D_{\max} = 1 - t_{\text{OFF},\min} / T = 1 - 0.25 / 133 = 99.8\%$$

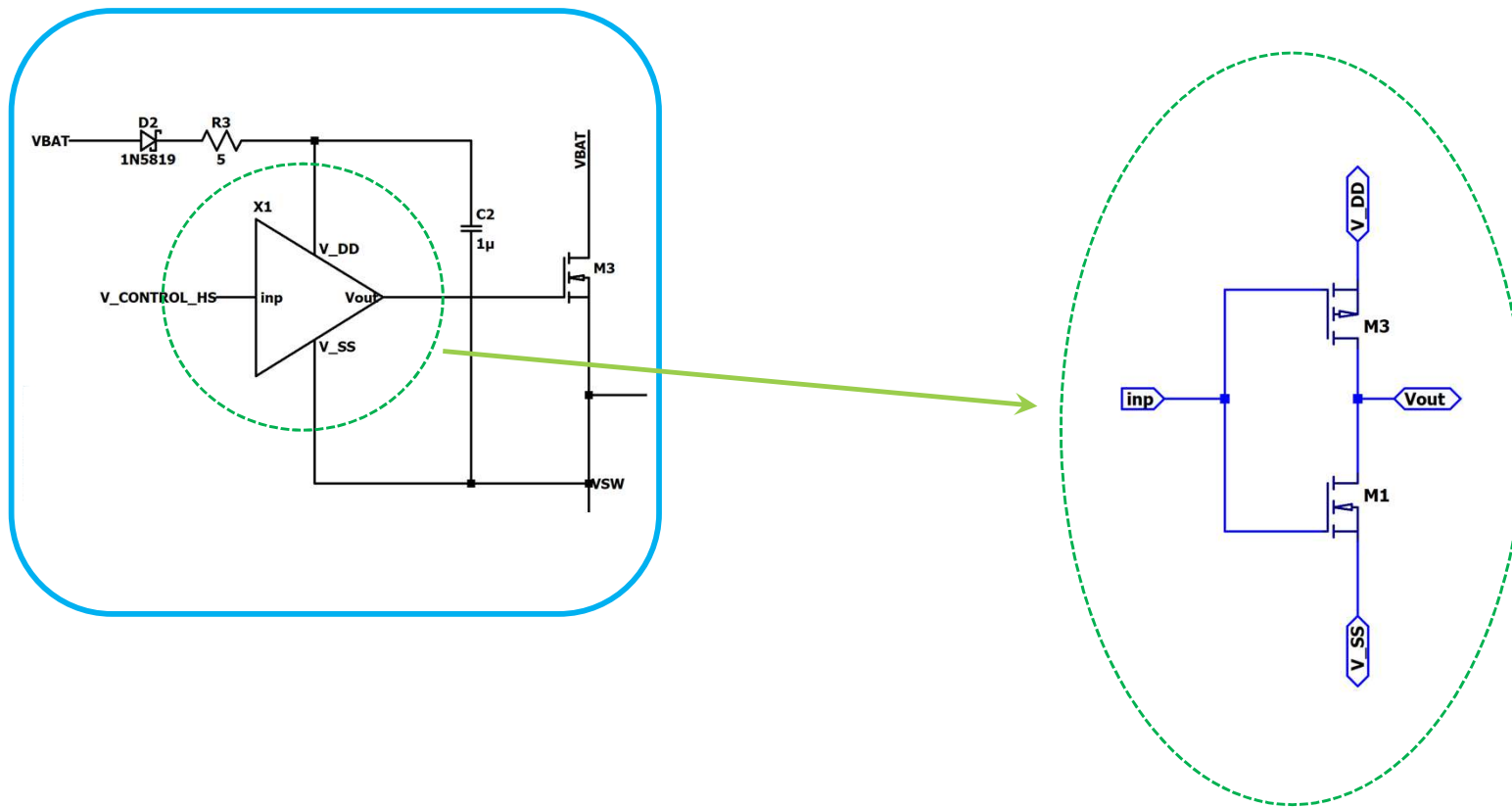
$$99.5\% > 97.5\%$$



$$D_{\max} = 130 \mu\text{s} / 133.33 \mu\text{s} = 97.5\% \rightarrow \text{Final safe duty cycle due to maximum allowable bootstrap capacitor voltage}$$

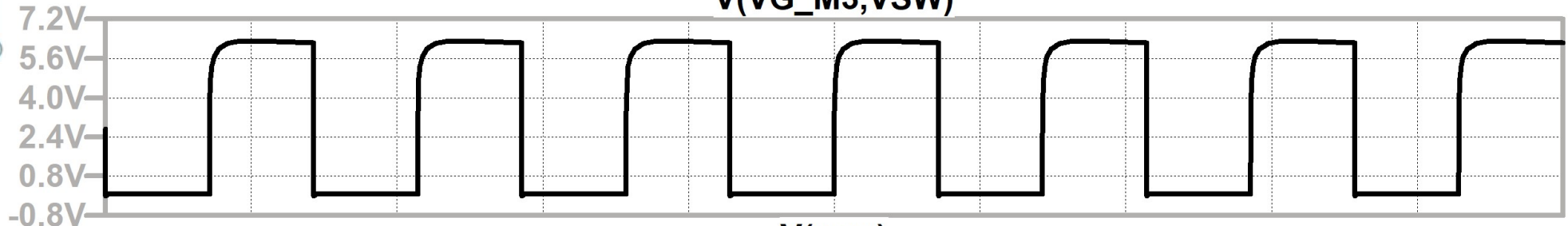
Bootstrap Circuits

Internal Structure of Buffer: Buffer circuit does not consume much power, since it just drives the gate of M3 MOSFET, therefore we can use a pmos/nmos pair.

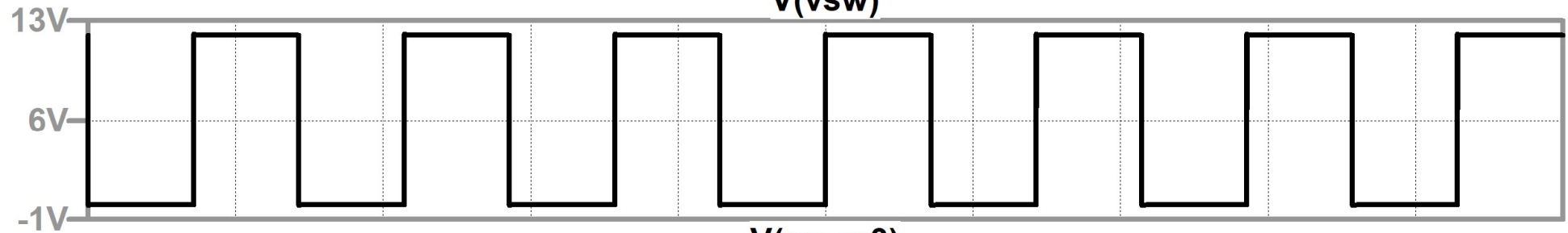


Bootstrap Circuits

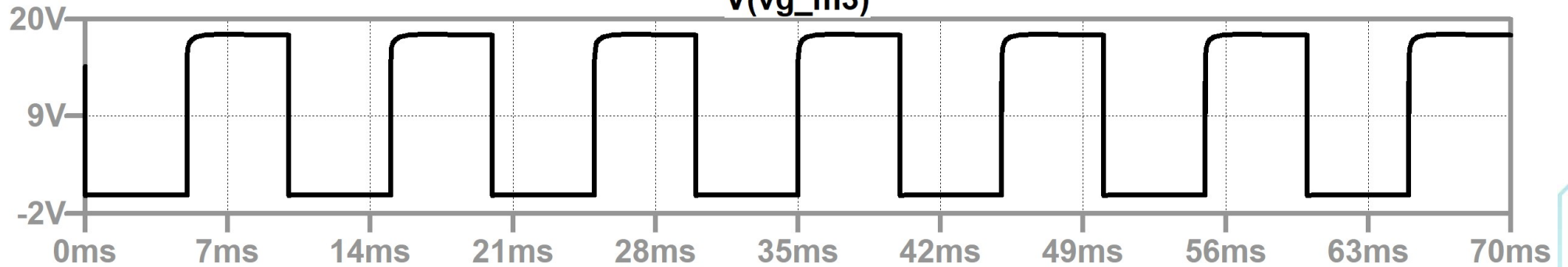
V(VG_M3,VSW)



V(vsw)



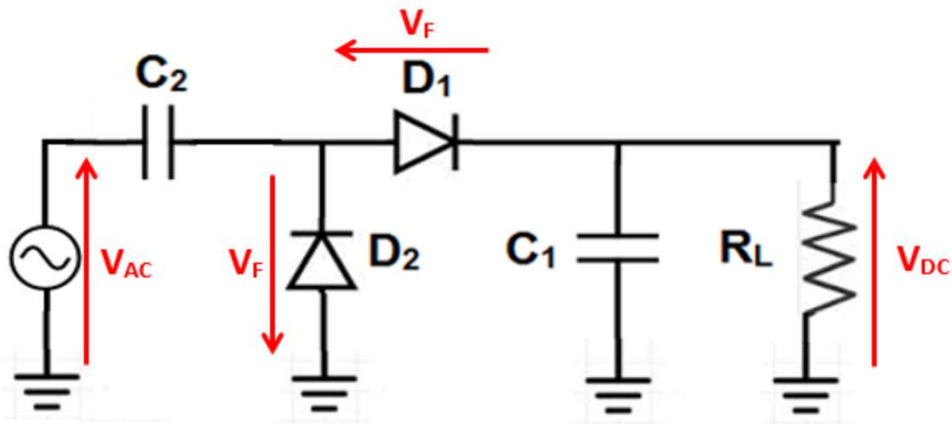
V(vg_m3)



Charge-Pump Circuits

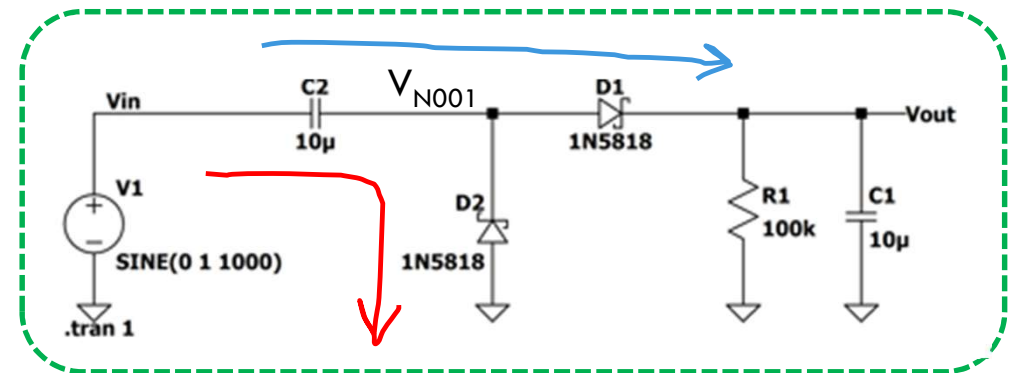
The voltage double circuit has already been covered in the previous weeks

$$V_{\text{out}} = V_{C2} + V_{\text{in,peak}} \longrightarrow \text{Diode voltage drop neglected}$$



Voltage Doubler Circuit

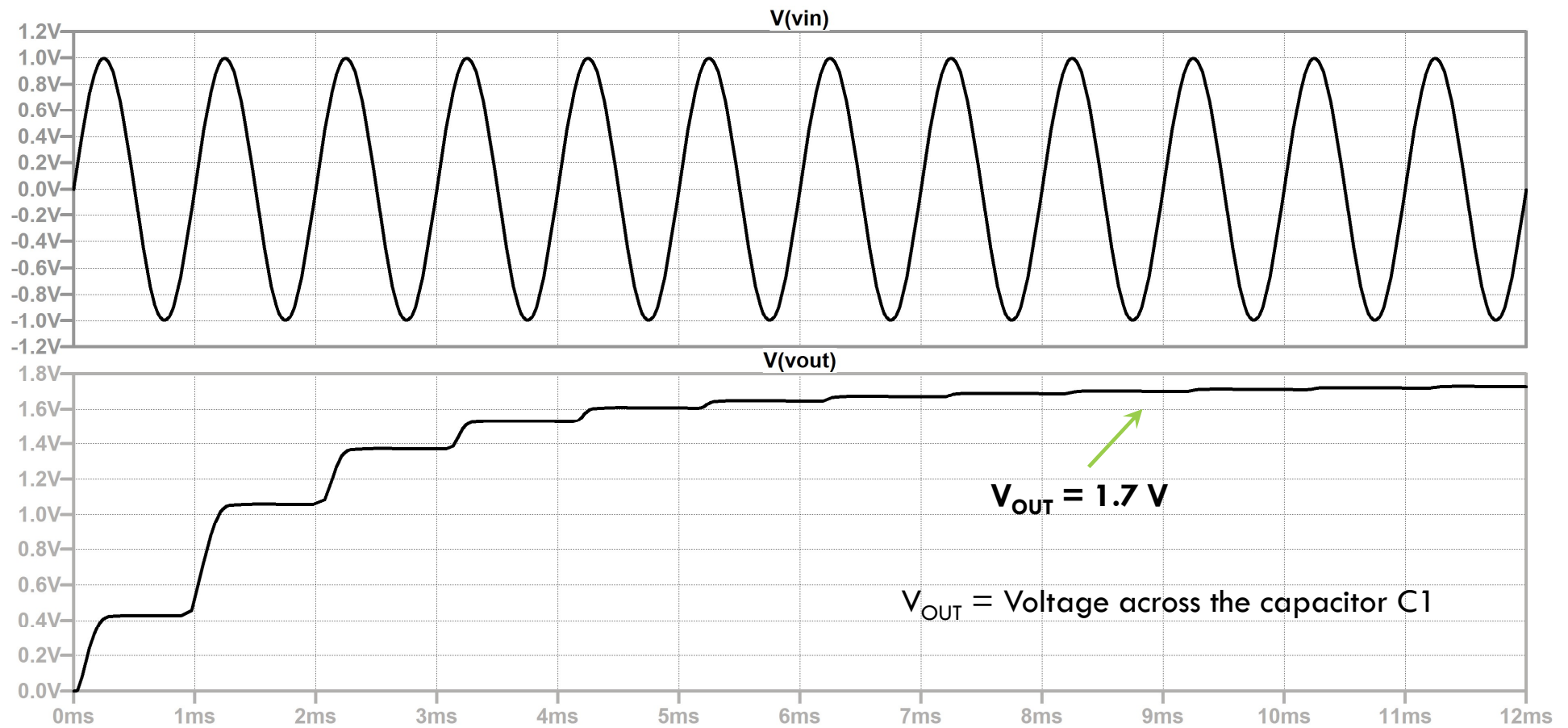
C_2 charged gradually both in negative and positive half-cycles!



LTSpice Simulation Circuit

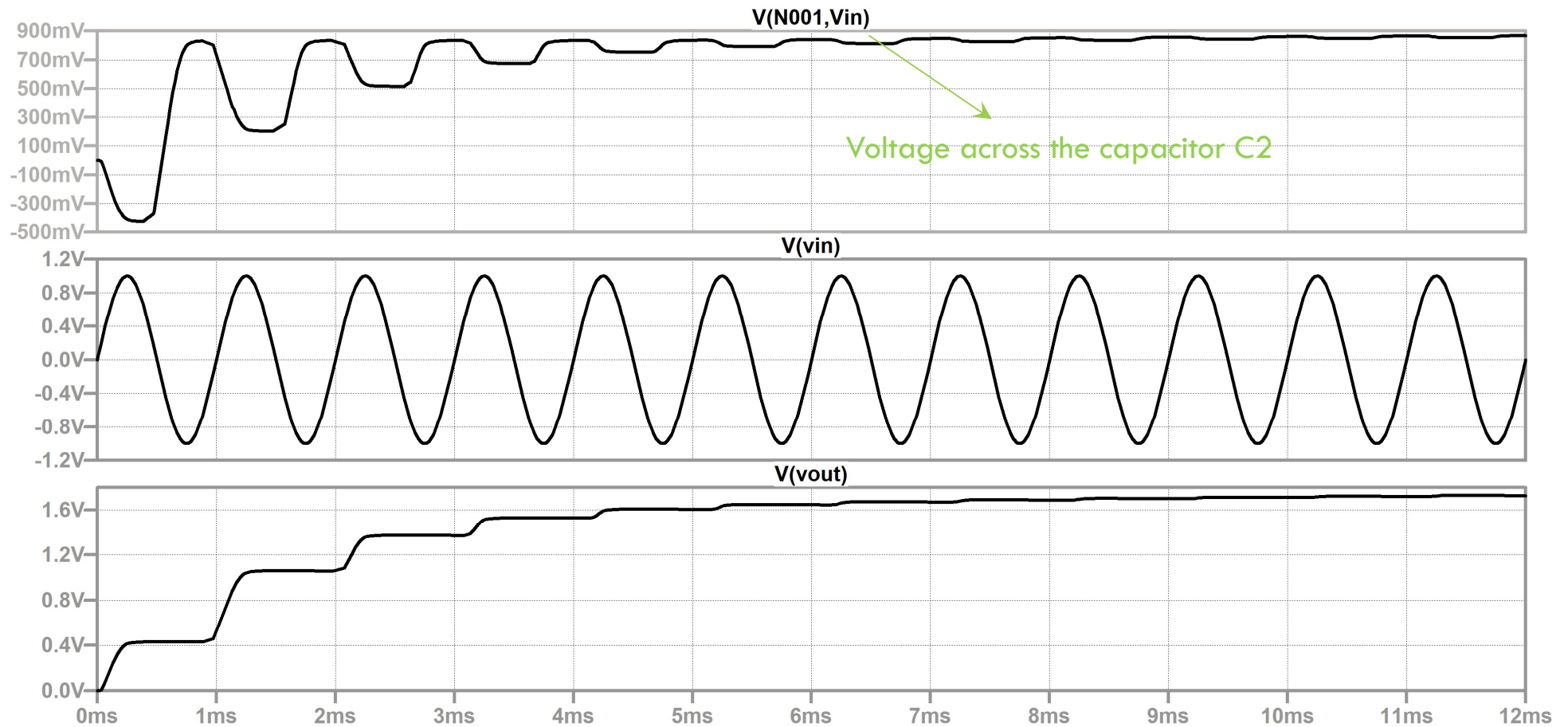
- : Current flow during negative V_{in}
- : Current flow during positive V_{in}

Charge-Pump Circuits



V_{OUT} smaller than 2 V due to the voltage drop across the diodes!

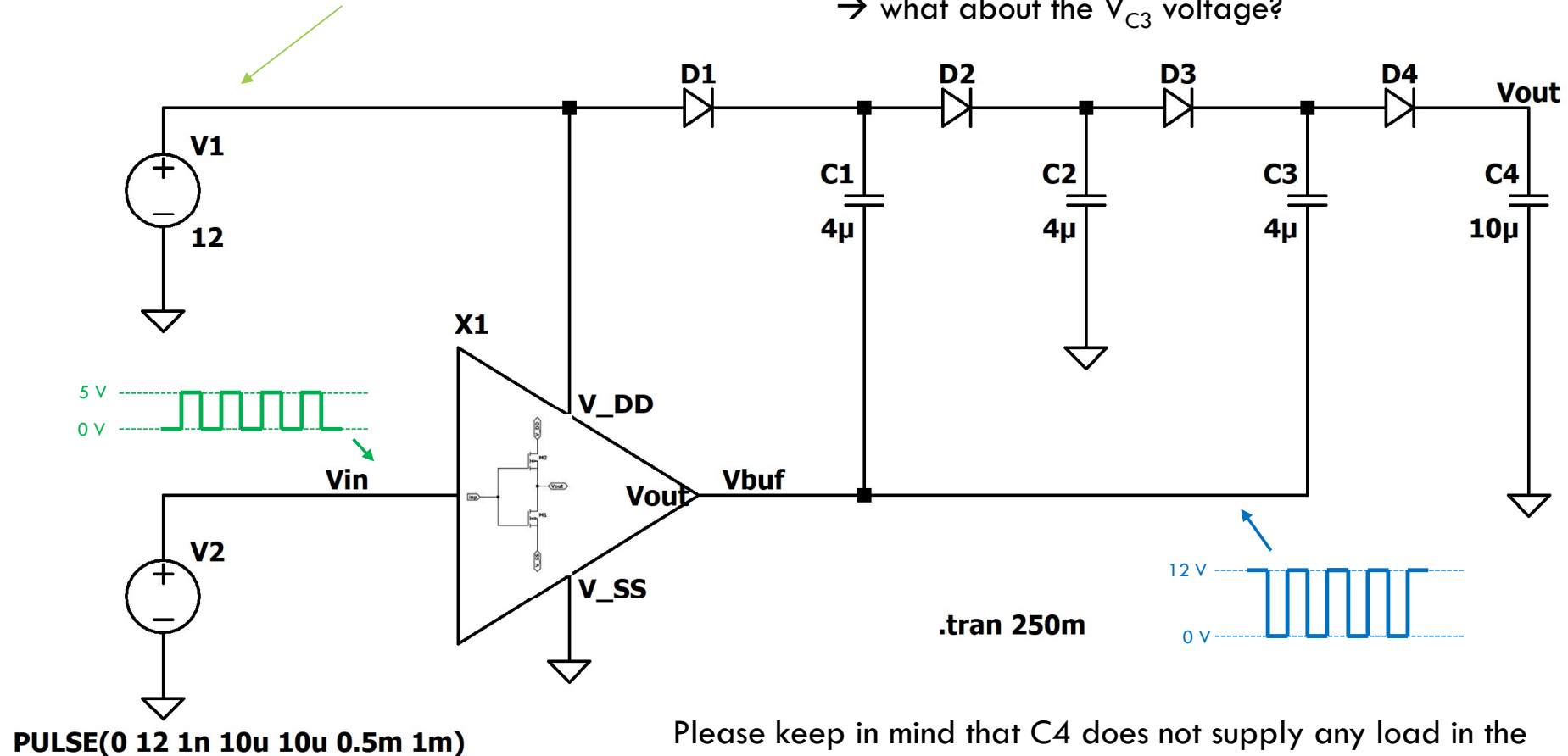
Charge-Pump Circuits



Charge-Pump Circuits

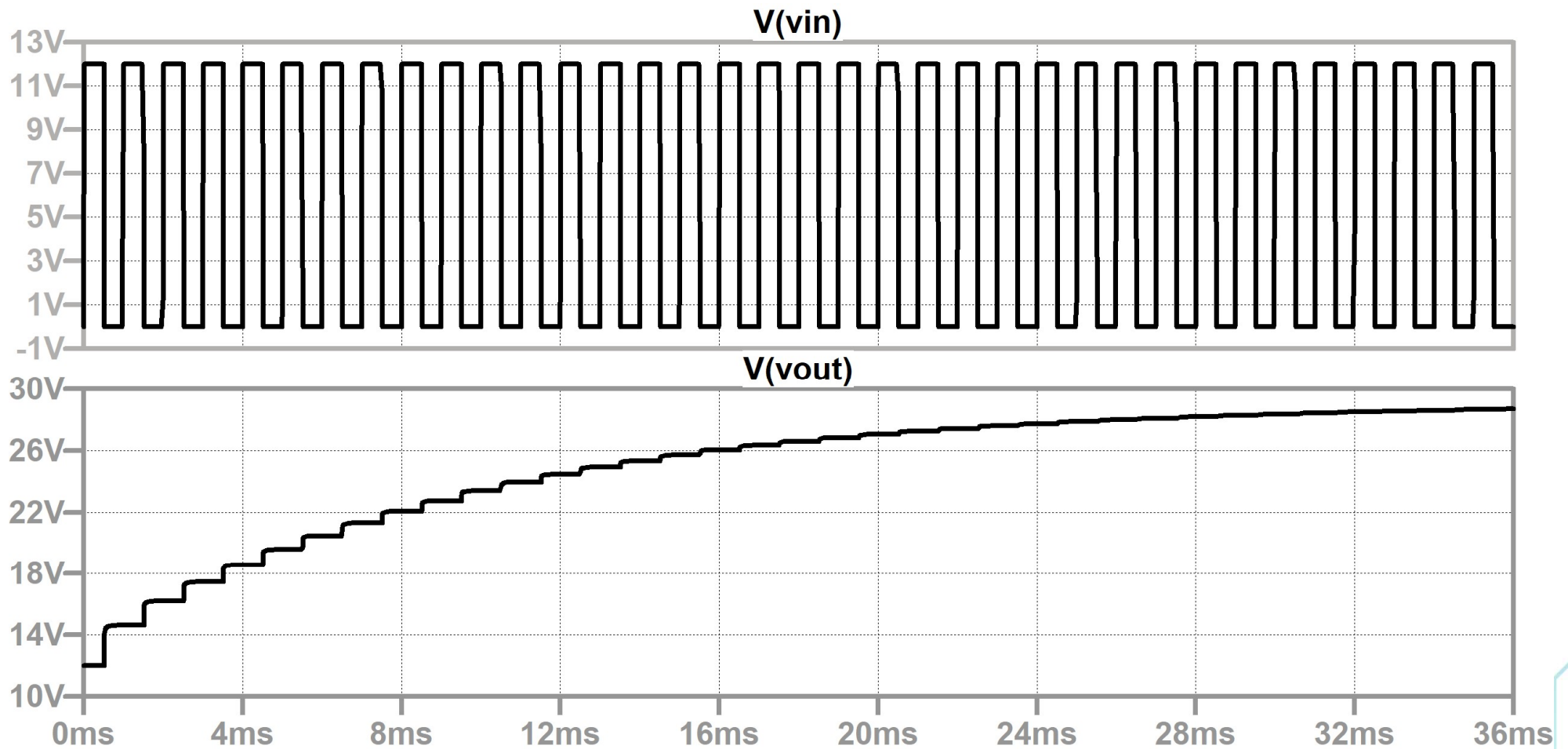
If the input voltage is not AC but DC....

If we neglect the voltage drop across the diodes $\rightarrow$ if $V_{in} = 5\text{ V}$, then $V_{C1} \approx 12\text{ V}$,
 $\rightarrow$ Next Cycle, if $V_{in} = 0\text{ V}$, $V_{C1} \approx 24\text{ V}$ and so on...
 $\rightarrow$ what about the V_{C3} voltage?

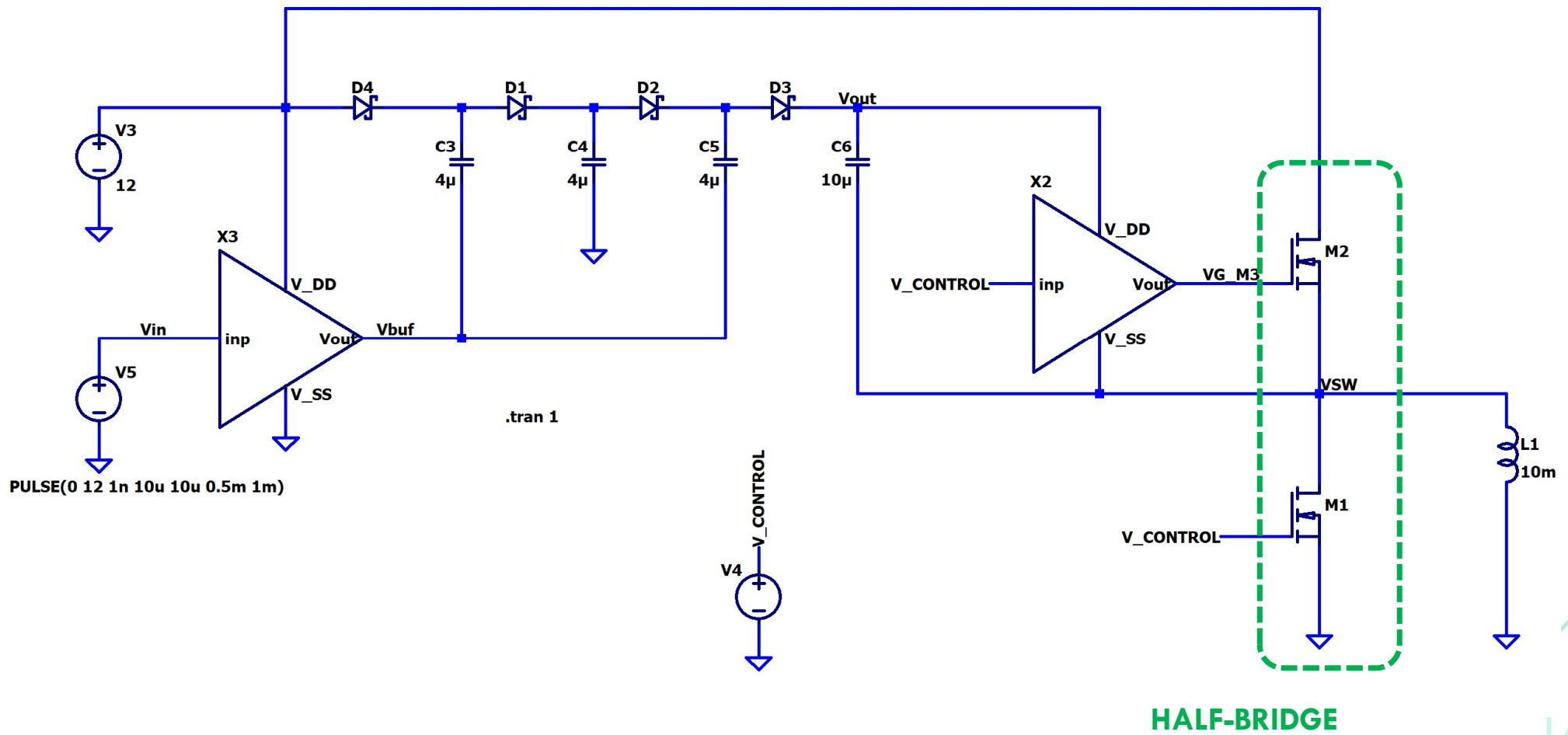


Please keep in mind that C4 does not supply any load in the simplified circuit above. If C4 supplies the load (e.g., buffer), C4 will be discharged and the voltage across C4 will be lower.

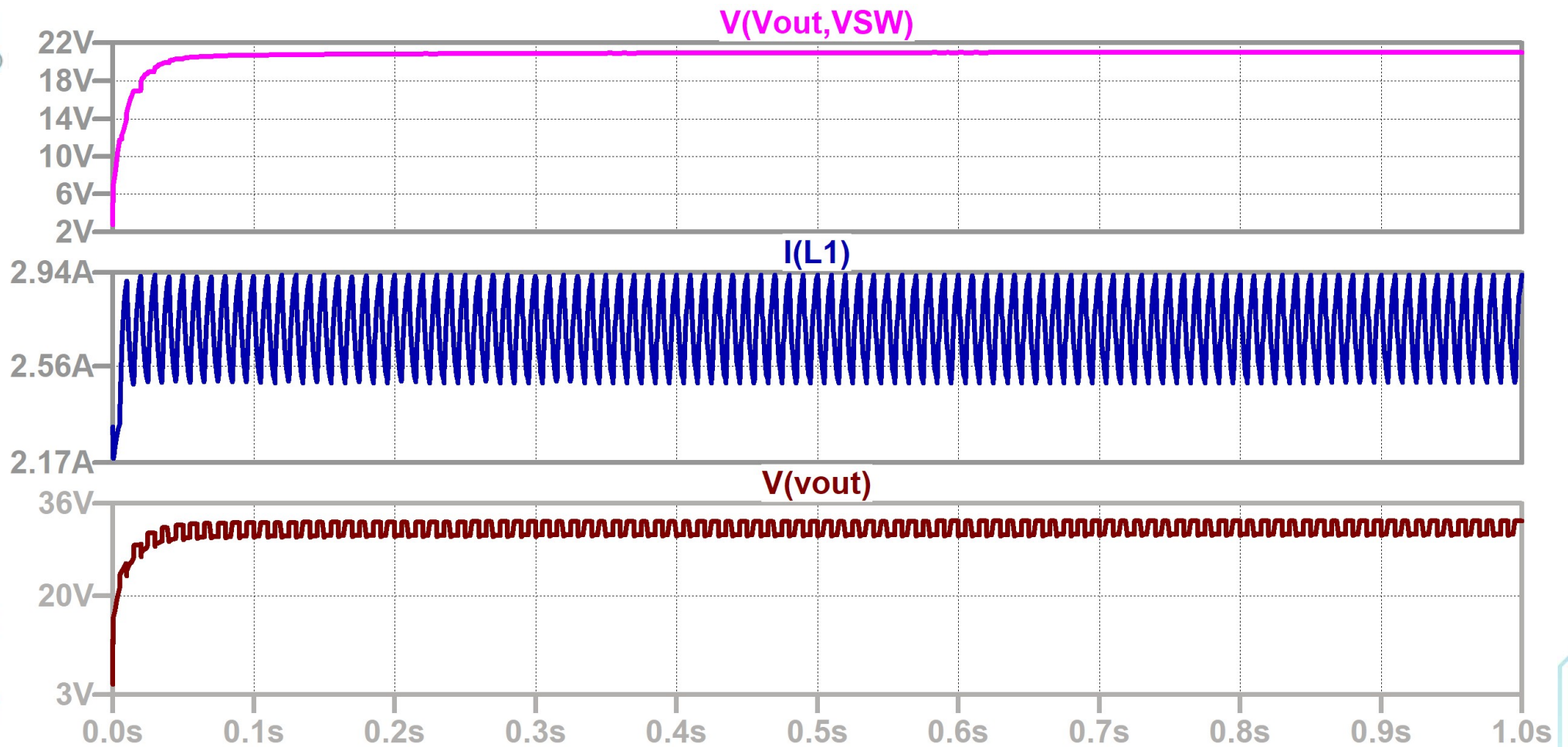
Charge-Pump Circuits



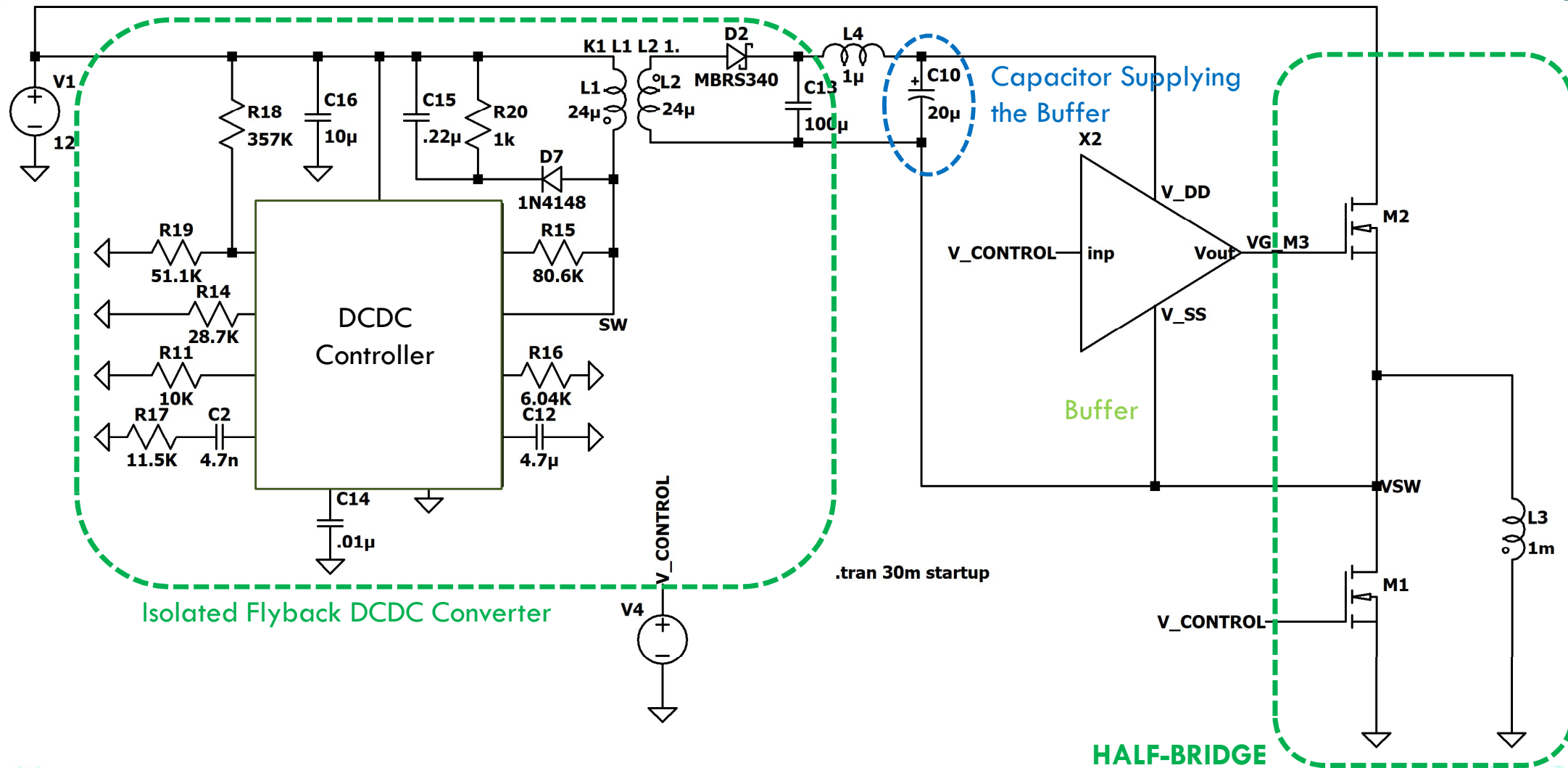
Charge-Pump Circuits



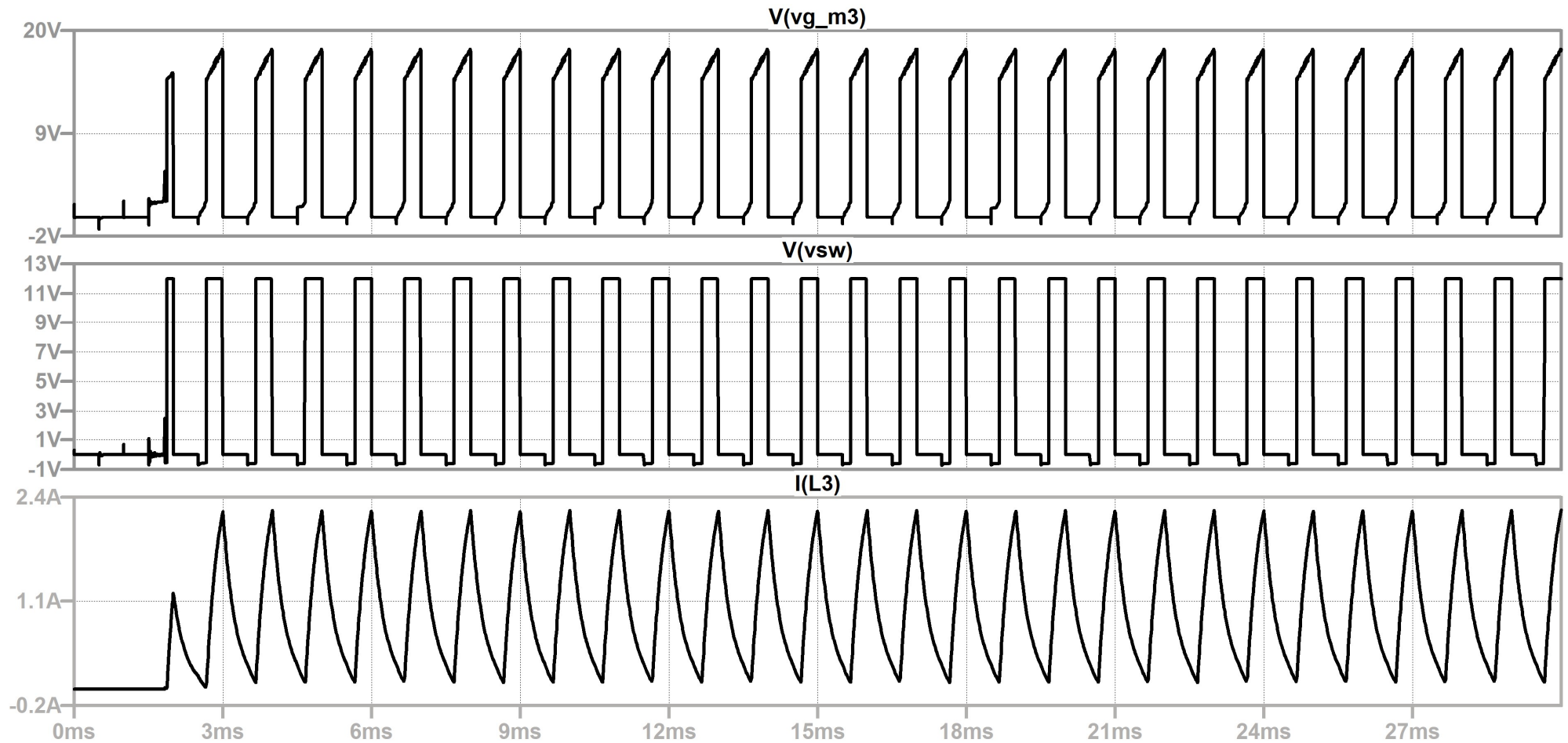
Charge-Pump Circuits



HALF-BRIDGE SUPPLIED BY ISOLATED DCDC CONVERTER



HALF-BRIDGE SUPPLIED BY ISOLATED DCDC CONVERTER

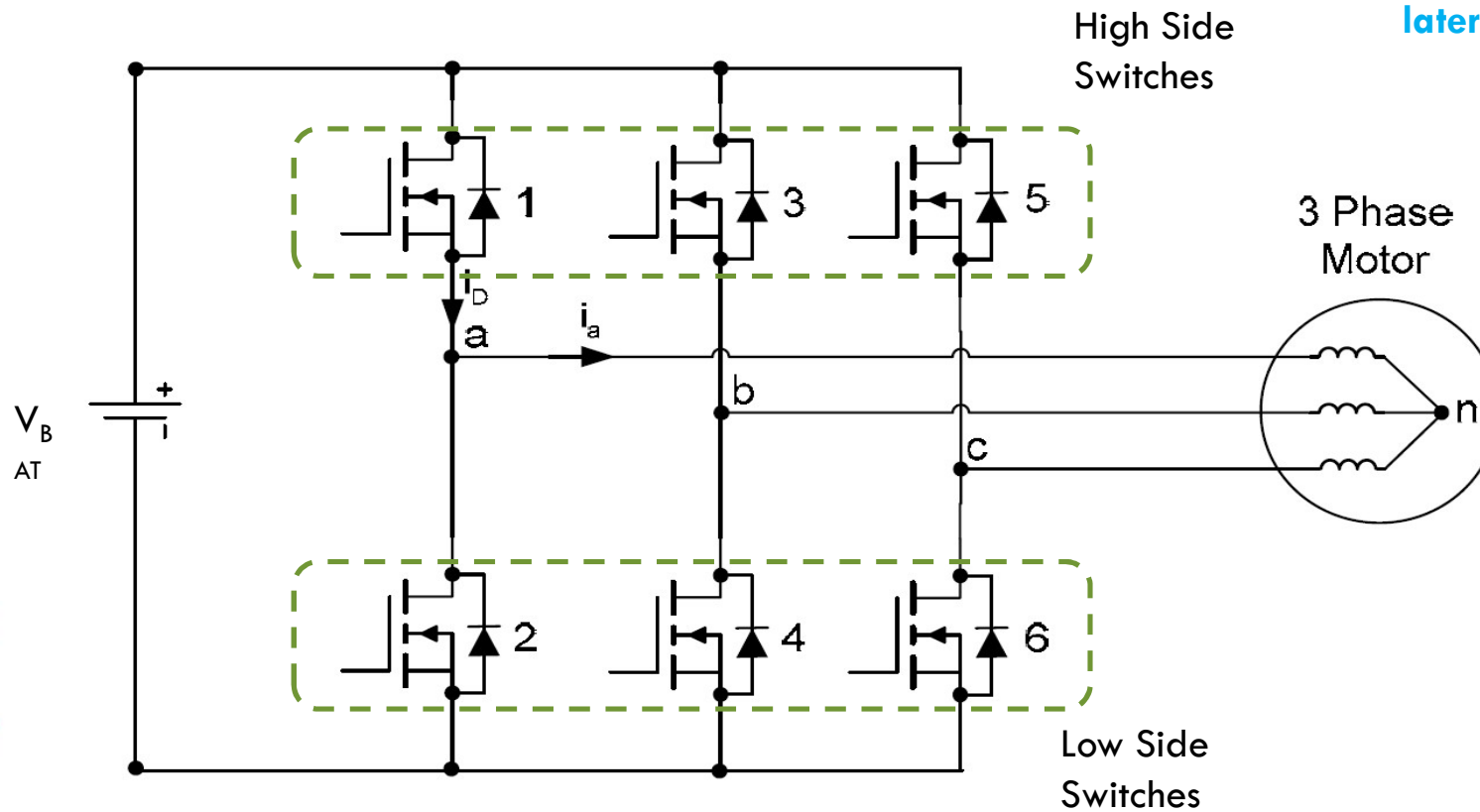


HALF-BRIDGE

Where do we use half-bridge circuits apart from DCDC converters?

Three-Phase Electric Motor Drives

This topic will be covered in the later weeks!

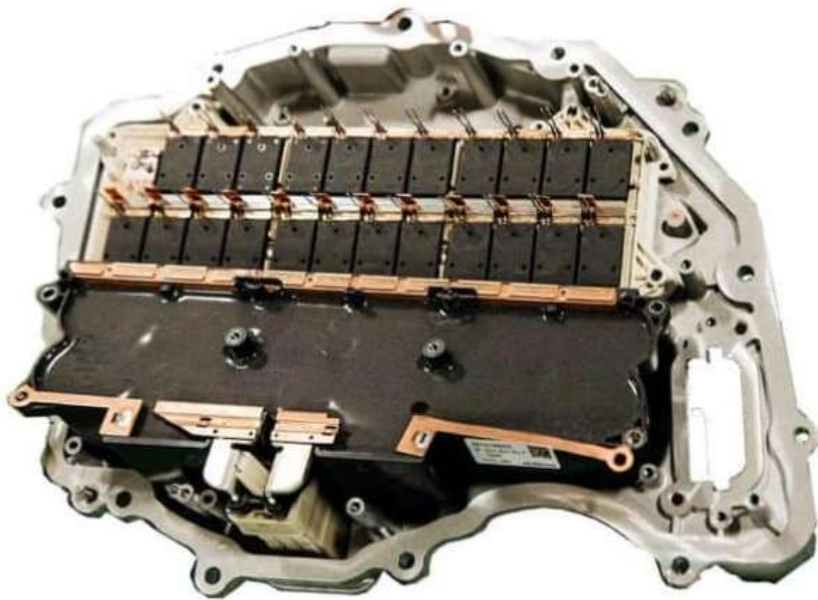


Usually PMSM and Induction Motors

HALF-BRIDGE

Where do we use half-bridge circuits?

Electric Motor Drives



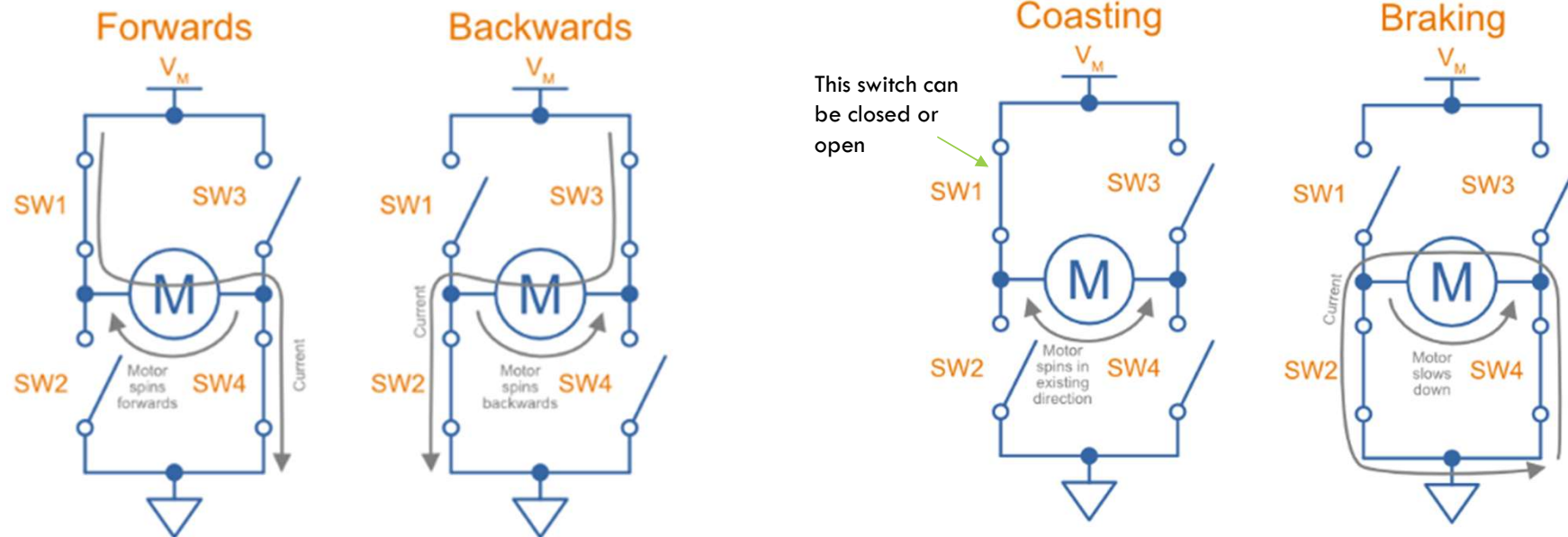
Tesla Model 3 Inverter

HALF-BRIDGE

Where do we use half-bridge circuits?

H-Bridge Electric Motor Drives

Charge-pump or bootstrap circuits necessary to drive n-channel high-side MOSFETs!!! Alternative: use p-channel MOSFETs → then no charge-pump etc. needed but pmos has a higher loss!



- ❖ Speed of the motor can be changed by adjusting the voltage across the motor
- ❖ Duty Cycle of the PWM signal applied to the gates of the MOSFETs change the voltage across the DC motor!

HALF-BRIDGE

Where do we use half-bridge circuits?

H-Bridge Electric Motor Drives

- Electric Throttle Valves
- Window Lift
- Sunroof
- Electrical Fan
- Electrical Pump



Window Lift Drive

HALF-BRIDGE

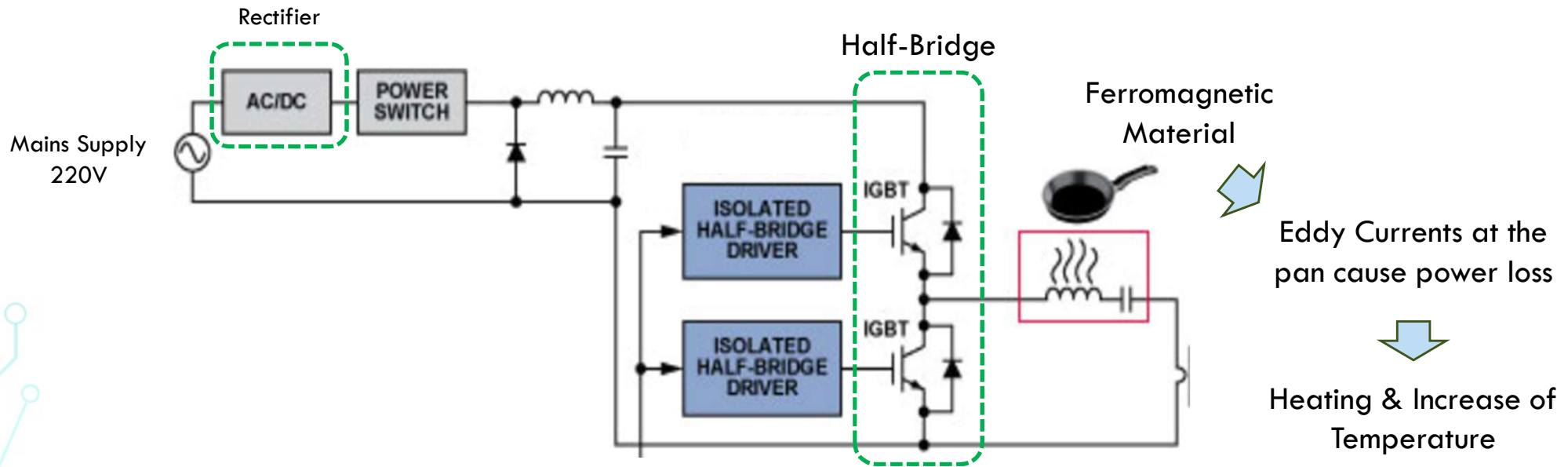
Where do we use half-bridge circuits?

Induction Cooking



Vestel AO-6470 S-K 60 cm 4 Gözü Elektrikli Ankastre Ocak

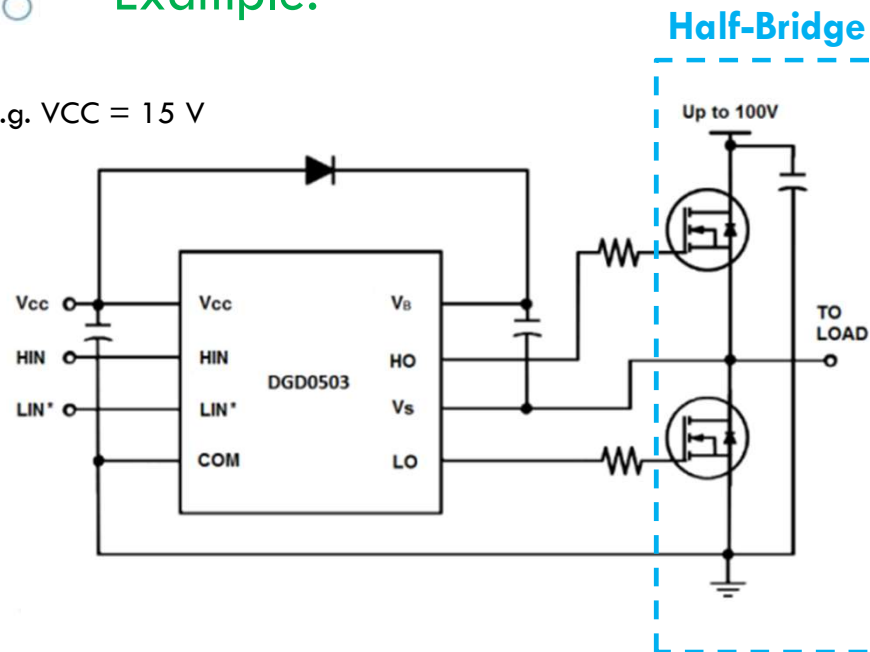
- Indüksiyon
- 4 Gözü Elektrikli
- 60 cm Genişlik
- Kayar Dokunmatik Kontrol
- Çocuk Kiliti



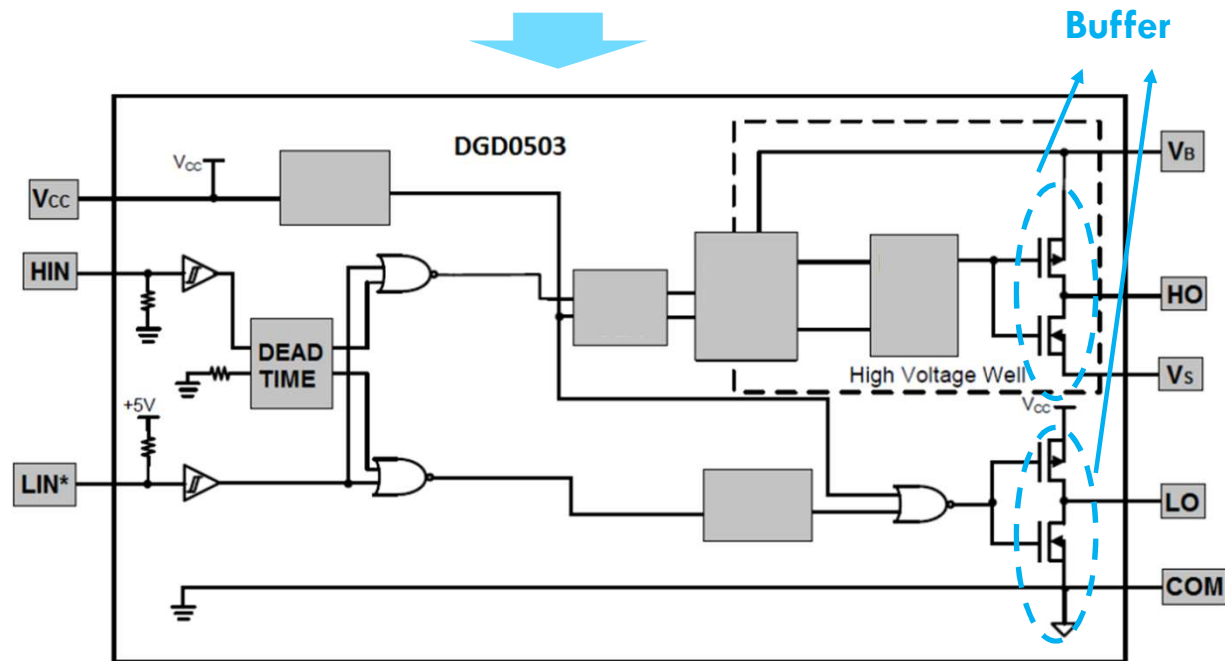
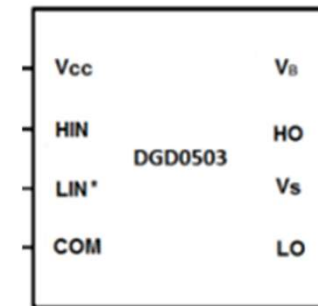
GATE DRIVERS

Example:

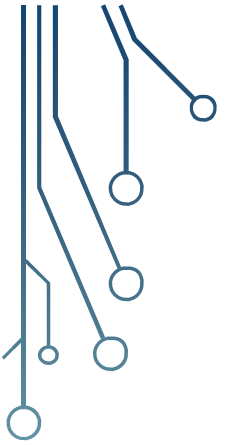
e.g. $V_{CC} = 15\text{ V}$



Simplified **Bootstrap** Gate Driver
Circuit with DGD0503



Inner Structure of DGD0503 Gate Driver IC



References

- Ribbens, William B. *Understanding automotive electronics* / William B. Ribbens; with contributions to previous editions by Norman P. Mansour ... [et al.] – 6th ed..
 - Analysis and Design of Analog Integrated Circuits”, Gray Mayer et al, 5th Edition, Wiley, 2009
 - Power Topologies Handbook, Markus Zehendner, Matthias Ulmann, Texas Instruments, 2020
 - Design and Application Guide of Bootstrap Circuit for High-Voltage Gate-Drive IC, AND9674/D, On Semiconductor, Application Note, 2021
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